Lab 1: RedFlag Descriptive Paper

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1 Introduction

The rise of digital communication and social media platforms has significantly changed how people consume information. News, opinions, and current world events can now be shared instantly across platforms such as Instagram, X (formerly Twitter), and online news websites. While this accessibility has increased the speed of information distribution, it's also created an environment where misinformation can spread rapidly and influence public understanding. Misinformation refers to false or misleading information that may be unintentionally or intentionally shared, often without verification. This issue has become a growing societal problem because users are constantly exposed to large volumes of content and may not have the time or resources to verify every claim they encounter.

The spread of misinformation has been studied extensively in recent years, particularly on social media platforms. Research conducted by Massachusetts Institute of Technology found that false news spreads significantly faster and farther than truthful news online (Vosoughi et al., 2018). The study analyzed over 126,000 stories shared by approximately 3 million users and concluded that false information was 70 percent more likely to be reshared than truthful information (Vosoughi et al., 2018). Additionally, truthful stories took approximately six times longer to reach the same audience size compared to false stories (Vosoughi et al., 2018). These findings demonstrate how misinformation can gain momentum concerningly quickly, making it difficult for users to identify credible information before it spreads widely.

Current solutions for combating misinformation are often limited in accessibility and efficiency. Most fact-checking systems require users to manually search for information using external websites such as Snopes or PolitiFact. While these resources are effective, they interrupt the user's browsing experience and create friction that discourages consistent use. In many cases, users must leave the content they were viewing, search for verification, compare sources, and then interpret findings independently.

Another limitation in current misinformation solutions is the lack of contextual bias awareness. Political and social issues are often presented through biased viewpoints, reinforcing echo chambers where users are exposed to viewpoints that are similar to their own. This ends up creating filter bubbles, which limit critical analysis and balanced understanding of important issues (Pariser, 2011). Existing systems generally focus on fact verification but don't provide users with alternative perspectives that may improve informed decision making.

[Figure 1: Graph showing misinformation growth]

To address these gaps, a solution must be passive and immediate, integrated directly into the user's browsing experience. RedFlag is designed as a real-time misinformation detection and bias awareness tool that addresses these needs. It operates as a background browser extension that scans content as users browse the internet, identifies potentially misleading information, and flags it on the screen. It also introduces BiasStar, a balanced perspectives engine that identifies politically divisive content and presents sources from both sides of the political spectrum. By combining real-time fact checking and bias awareness, RedFlag aims to improve and support more informed decision-making in online environments.

2 Product Description

RedFlag is a fact-checking and bias indicator utility that works across multiple platforms. It runs in the background and scans text to find any inaccuracies and provide sources for the user to look into. This information and sources provided give the user a way to look further into the misinformation. There is also the BiasStar which will add a star to any politically divisive content and offers further viewpoints from sources across the political spectrum.

[Figure 2: RedFlag High Level Concept Diagram]

2.1 Key Product Features and Capabilities

RedFlag comes with several features and capabilities to set it apart from other fact-checking tools. First, it comes with real-time content detection, it scans the user's screen to find any misinformation, working automatically in the background without requiring any user input. It addresses the inconvenience of the user having to leave their current browsing experience by bringing sources directly to them.

Next, it can highlight any questionable information on the users' browsing experience with no input, not intruding on the users' experience and presents the highlight subtly. It intends to intervene before the user potentially shares misinformation, whilst not entirely interrupting the experience.

RedFlag also comes with a balanced perspective engine, called BiasStar, that identifies political content and provides multiple viewpoint sources across the political spectrum. It does not label ideologies explicitly but does offer an evaluation from every perspective. It addresses how social media may reinforce bias and echo chambers.

RedFlag users can also get on-demand explanations on why the information they were seeing was flagged. This feature allows the user to make a more informed decision without forcing how the user should think about the topic, providing better credibility towards content and allowing the user to think further on the information provided.

2.2 Major Components (Hardware/Software)

[Figure 3: RedFlag MFCD]

RedFlag's presentation layer consists of the RedFlag browser extension and is both the front-end interface, and is responsible for displaying the real-time alerts and a view panel to users. This layer provides the primary interface through which users interact with RedFlag whilst browsing. It's built with JavaScript, allowing it to run on browsers with ease.

RedFlag's application layer contains the primary processing components of RedFlag, including content detection, misinformation analysis, fact verification, the Alert and Recommendation generator and the Balanced Perspectives Engine. These components work all together to analyze content and determine what information should be presented to the user. The backend uses Python with Flash/Django frameworks to process information and interact with the project's information datasets.

The data layer consists of a PostgreSQL database which is used to store the information required by RedFlag's various systems. It includes the Source Credibility Database, which is responsible for containing the reliability scores and political alignments, as well as User Interaction data and Content Intelligence data. The Content Intelligence data keeps the known false claims and topic classifications, allowing the application layer to compare newly detected information against previous information data.

Finally, RedFlag uses various External Interfaces and APIs to supplement its various systems. The fact-checking API compares questionable information against any available databases, while the News API takes publicly available news as input then categorizes it by topic. An LLM API is used to parse database information and summarize its findings, which in turn helps increase the speed at which RedFlag can perform fact-checking and present information to the user.

3 Identification of Case Study

Daiskue is a 16-year old teenager who spends a large amount of his free time scrolling through Instagram on his phone. Although he recognizes that some of the information appearing on his Instagram feed may be false, he does not want to spend the time researching every post he encounters. RedFlag is being developed to address this problem by providing misinformation indicators that allow users to identify potentially questionable information without having to independently research every post. Daiskue enables RedFlag through his browser, which causes indicators to appear whenever questionable text is detected. RedFlag provides multiple sources of information related to the questionable content, allowing Daiskue to access those sources immediately and make a more informed decision about the information he encounters. In the future, RedFlag could be used by his friends and others who learn about RedFlag through word of mouth.

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4 Glossary

- **Misinformation: False or misleading information shared regardless of intent**
- **Disinformation: False or inaccurate information that is shared with the intention to mislead others**
- **End-user: The person who uses or is intended to use a computer system or other software application**
- **Credibility Score: A score given to a claim that is found online rating the credibility of the claim, or the likelihood that the claim is correct**
- **BiasStar: RedFlag's internal system for identifying politically divisive content**

5 References

Vosoughi, S., Roy, D.., & Aral, S. (2018). The spread of true and false news online https://www.science.org/doi/10.1126/science.aap9559

Pariser, E. (2011). *The filter bubble: What the Internet is hiding from you*. Penguin Press.